Networking:-

OSI and TCP/IP Models are standard Rules for sending data over a network.

when we send a message, stream a video, or open a website, data does not travel magically. it is broken down into small pieces stamped with source and destination addresses, converted into electrical or radio signals, and sent across cables or airwaves. These steps happen across structured layers

~> OSI Model (open systems interconnections)

The OSI model consists of 7 layers, it is a conceptual ~~tage~~ framework designed to understand how network communication work.

when sending data, it moves down from layer 7 to layer 1, when receiving data, it moves up from layer 1 to layer 7.

Layer 7:- Application layer (user Interface)

o function:- Provides network access directly to software applications.

o protocols:- HTTP, HTTPS, FTP, SMTP. (HyperText Transfer protocol secure) and SMTP (Simple Mail Transfer protocol).

o ex:- Opening a web page in Google chrome or sending an email via ~~Out~~ Out Look.

Layer 6:- Presentation Layer: (Translator and Security)

Function:- Translates data formats, handles encryption/decryption and compressess data.

ex:- Encrypting credit card details into secure cipher text before sending, or formatting a file as a PNG Image.

Layer 5: Session layer (connection controller)
- function:-
Establishes, maintains, and terminates connections (sessions) b/w two devices
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Ex:- Getting automatically logged out of a online banking portal after 5 minutes of inactivity

Layer 4:- Transport Layer. (End-to-End Delivery and Reliability)
- function: splits data into smaller segments, assigns port numbers, and ensures complete data delivery.

- protocols: TCP (Reliable), UDP (Fast)

→ TCP and UDP are Two primary protocols used to send data packets across the internet.
The operate at the Transport layer of the networking model, but they prioritize entirely different Goals.
TCP focuses on absolute accuracy, while UDP focuses on maximum speed.
ex:- A shipping service attaching tracking numbers to ensure no packages are lost in transit.

• Layer 3:-
Network Layer Routing:- (Routing and Logical Addressing)
• function:- Adds IP addresses to data [(creating packets) and determines the best physical route across networks.

• key Device:- Router

ex:- writing the recipents street and postal code on an envelope.

Layer 2:- Data Link Layer (Physical Addressing and error Detection.
function:- Encapsulates packets into frames, handles hardware (MAC) addressing, and manage local network transmission.
Key Device:- Network switch.

ex:- Using your laptops unique factory assigned Wifi MAC adress to connect to a local router.

Layer 1:- Physical layer (Hardware and Signal Transmission)

- function: converts raw binary data (0s and 1s) into physical signals (electrical, optical, or radio waves).

- Key Devices:- Ethernet, cables, Fiber optics, WiFi Antennas

ex/. physical copper wires carrying electrical voltage across a local are network.